

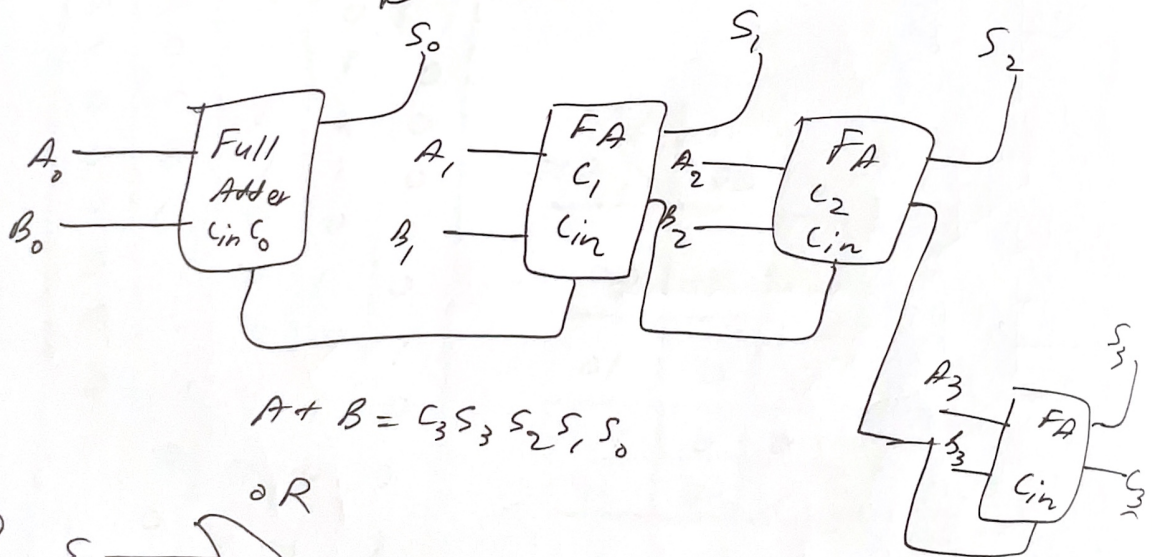
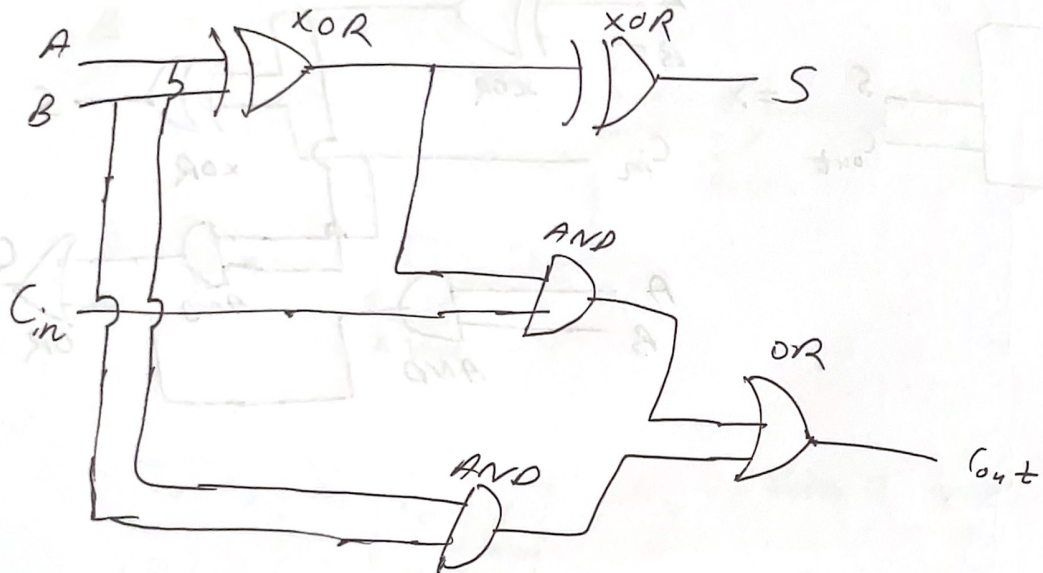
EE214 Lab-4 Report

Name: Medhansh

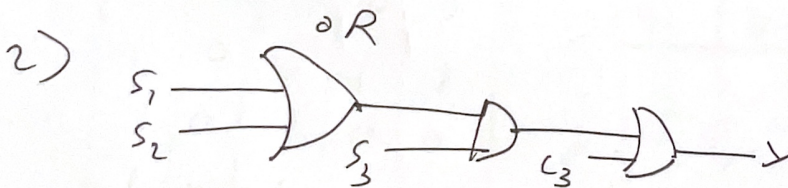
Roll No.: 25B3986

Name: Medhansh
Roll No.: 25B3986

1) Full Adder



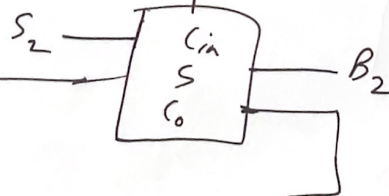
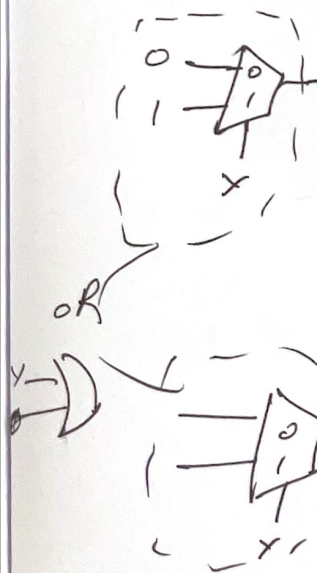
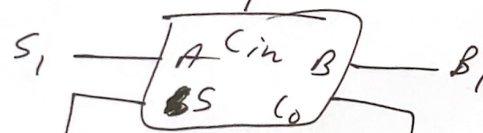
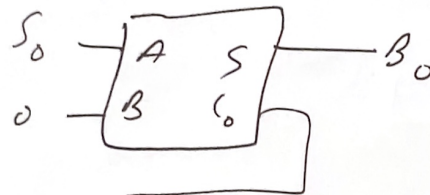
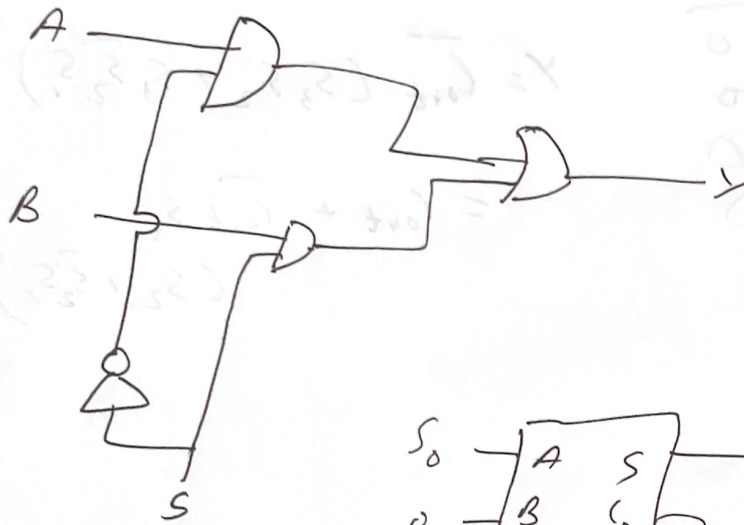
$$A + B = C_3 S_3 S_2 S_1 S_0$$



$$Y = 1 \quad \text{sum} > 9$$

$$Y = 0 \quad \text{sum} \leq 9$$

3) 2-bit mux



Truth Table / K-Map

	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	1	1	1
10	0	0	1	1

$$Y = \overline{C_{out}} (S_3 S_2 + S_3 \overline{S_2} S_1)$$

$$= \overline{C_{out}} + \overline{C_{out}} S_3 (S_2 + \overline{S_2} S_1)$$

Verified
Dutthap
28/8/26

BCD Adder

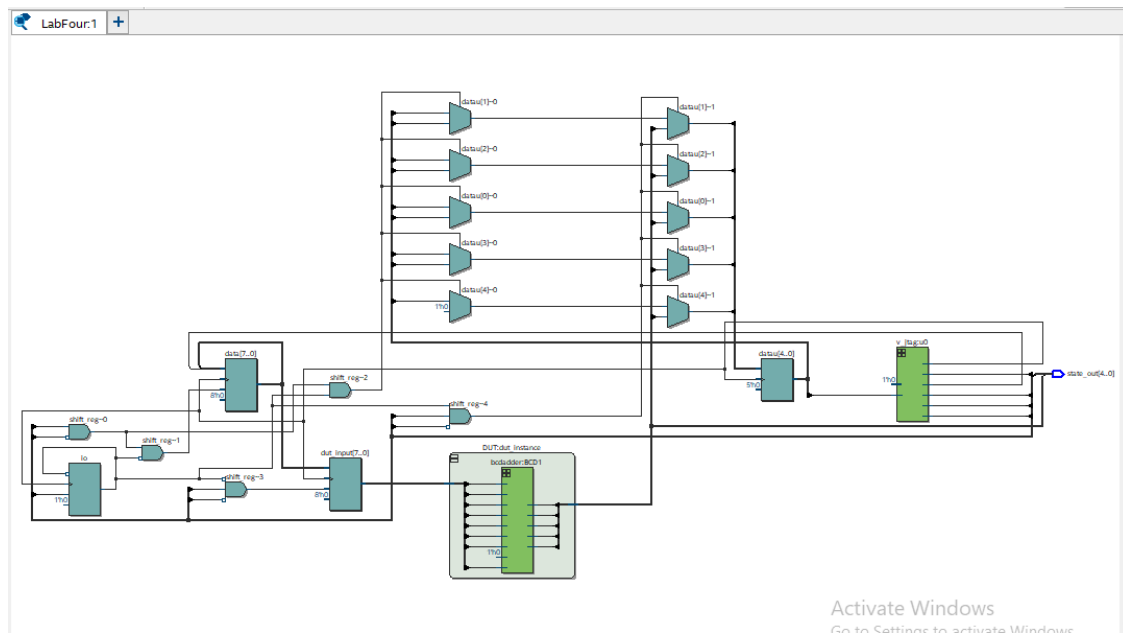


Figure 1: Schematic

```
00000000 11111 Success
00000001 11111 Success
00000010 11111 Success
00000011 11111 Success
00000100 11111 Success
00000101 11111 Success
00000110 11111 Success
00000111 11111 Success
00001000 11111 Success
00001001 11111 Success
00010000 11111 Success
00010001 11111 Success
00010010 11111 Success
00010011 11111 Success
00010100 11111 Success
00010101 11111 Success
00010110 11111 Success
00010111 11111 Success
00011000 11111 Success
00011001 11111 Success
00100000 11111 Success
00100001 11111 Success
00100010 11111 Success
00100011 11111 Success
00100100 11111 Success
00100101 11111 Success
00100110 11111 Success
00100111 11111 Success
00101000 11111 Success
00101001 11111 Success
00110000 11111 Success
00110001 11111 Success
00110010 11111 Success
00110011 11111 Success
```

Ln 4, Col 23 | 2,300 characters | Plain text

Figure 2: Output file

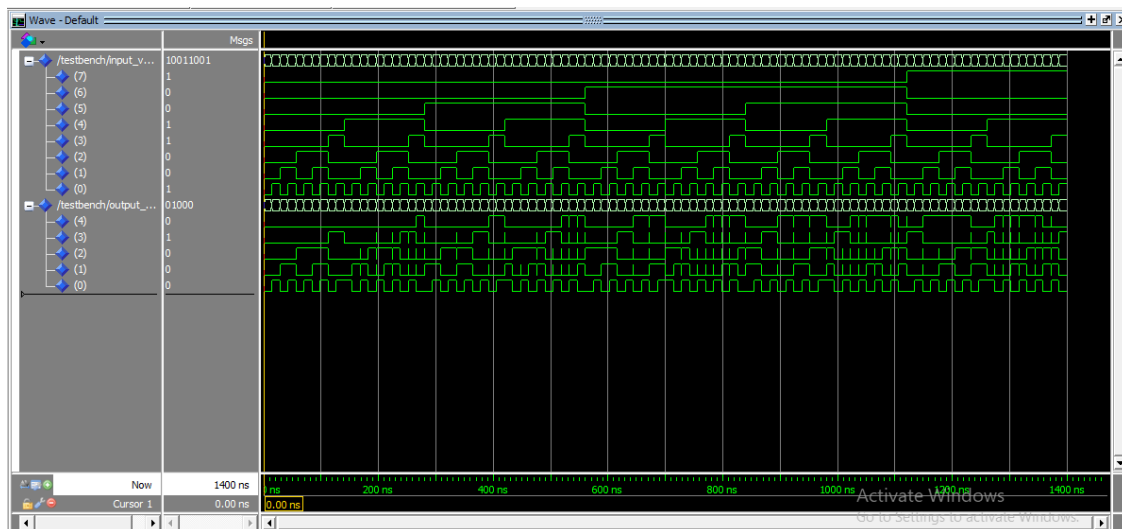


Figure 3: Waveform

```
#
# add wave *
# view structure
# .main_pane.structure.interior.cs.body.struct
# view signals
# .main_pane.objects.interior.cs.body.tree
# run -all
# ** Note: SUCCESS, all tests passed.
#   Time: 448 ns   Iteration: 0   Instance: /testbench

/SIM 2> ]
```

Figure 4: Transcript